e-PG Diploma AI & DS (AUG '25)

Statistical Foundations of Machine Learning

Quiz 2

November 15, 2025

Multiple Choice Questions (MCQs)

Question 1. Consider the following statements, where X, Y, Z are discrete random variables. Which of these statements are true?

(A) If X and Y are independent and Y and Z are independent, then X and Z are independent.

(B) If X and Y are independent, then they are conditionally independent given Z.

(C) If X and Y are conditionally independent given Z, then they are independent.

Choose the correct option:

(a) (A) and (B)

(b) (B) and (C)

(c) Only (A)

(d) **None of the above**

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Question 2. Four fair dice are rolled. Find the expected total of the rolls.

(a) 10

(b) 12

(c) **14**

(d) 16

Question 3. Let $X, Y \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,1)$. Consider:

(A) $X+Y$ and $X-Y$ are i.i.d. $\mathcal{N}(0,2)$.

(B) $\text{Cov}(X+Y, X-Y) = 0$.

(C) $X+Y$ is independent of $X-Y$.

Which of the above statements are true?

(a) (A) and (B)

(b) Only (A)

(c) (A) and (C)

(d) **(A), (B), and (C)**

Question 4. Which of the following statements about the Student-t distribution t_n are true?

(A) If $T \sim t_n$, then $-T \sim t_n$.

(B) As $n \to \infty$, the t_n distribution approaches the standard Normal distribution.

(a) Only (A)

(b) Only (B)

(c) **Both (A) and (B)**

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(d) None

Question 5. If $X \sim N(25, 16)$, what is the probability that X lies between 13 and 37?

(a) Approximately 68%

(b) Approximately 95%

(c) **Approximately 99.7%**

(d) Cannot be determined

Question 6. A **Type II error** occurs when:

(a) We reject the null hypothesis when it is true.

(b) **We fail to reject the null hypothesis when it is false.**

(c) We reject the alternative hypothesis when it is true.

(d) We accept the alternative hypothesis when it is false.

Question 7. Which of the following statements are true?

(A) A statistical hypothesis is a statement about the nature of a population.

(B) A test statistic is determined from the sample data.

(C) The critical region is the set of values of the test statistic for which the null hypothesis is accepted.

(a) **(A) and (B) only**

(b) (B) and (C) only

(c) (A) and (C) only

(d) (A), (B), and (C)

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Question 8. When the population standard deviation σ is known, the hypothesis about the population mean is tested by:

(a) t-test

(b) **Z-test**

(c) χ^2-test

(d) F-test

Question 9. As the sample size increases, the t distribution becomes more similar to the:

(a) χ^2 distribution

(b) Uniform distribution

(c) **Normal distribution**

(d) F distribution

Question 10. The p-value represents:

(a) **The smallest significance level at which H_0 can be rejected**

(b) The probability of Type II error

(c) The largest significance level at which H_0 can be rejected

(d) The probability of Type I error

Question 11. A professor sees students during office hours. Time spent follows an exponential distribution with mean 10 minutes. What is $P(X < 20)$?

(a) 0.1353

(b) 0.5

(c) **0.8647**

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(d) 0.9817

Question 12. The lives of spark plugs are $N(60{,}000, 4{,}000^2)$. A sample of 16 plugs has $\bar{X} = 58{,}500$. What is $P(\bar{X} \leq 58{,}500)$?

(a) **0.0668**

(b) 0.4332

(c) 0.9332

(d) 0.0175

Question 13. A café records daily customer counts: $12, 8, 15, 7, 9, 10, 6, 5, 14, 8$. Estimate the proportion of days with ≤ 8 customers. [Give marks for 0.5 and cut for 0.37]

(a) 0.25

(b) 0.37

(c) **0.50**

(d) 0.63

Question 14. A coating's thickness (mm) from 9 samples is given by

$$19.8,\ 21.2,\ 18.6,\ 20.4,\ 21.6,\ 19.8,\ 19.9,\ 20.3,\ 20.8.$$

Find a 90% confidence interval for the population variance (assume normality). [Give marks for everyone]

(a) **(0.406, 2.305)**

(b) (0.637, 1.518)

(c) (0.553, 1.122)

(d) (2.733, 15.507)

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Question 15. A company claims that its new battery lasts at least 10 hours on average. A consumer group tests this claim with the following hypotheses:

$$H_0 : \mu = 10 \quad vs \quad H_1 : \mu < 10 \tag{1}$$

Which of the following describes a Type I error in this context?

(a) Concluding that the average battery life is less than 10 hours when it actually is 10 hours or more.

(b) Concluding that the average battery life is 10 hours or more when it actually is less than 10 hours.

(c) Failing to test enough batteries to detect a difference from 10 hours.

(d) Using the wrong level of significance for the test.

Question 16. Which best distinguishes a sample from a population?

(a) A sample includes every member of a group, while a population includes only a few selected members.

(b) A sample is a subset of the population that is used to draw conclusions about the entire population.

(c) A population is always smaller than a sample.

(d) A population consists only of data collected from experiments, while a sample comes from surveys.

Question 17. Which of the following best describes the Central Limit Theorem?

(a) It states that the mean of a population is always normally distributed, regardless of the population's shape.

(b) It states that as the sample size increases, the sampling distribution of the sample mean approaches a normal distribution, regardless of the shape of the population.

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(c) It states that large samples always have the same mean as the population mean.

(d) It states that population data become normal when the population size is large enough.

Question 18. When attempting to land a drone on a target in two-dimensional space, suppose the horizontal and vertical positioning errors are independent normal random variables each with mean 0 and standard deviation 1.5 meters. Find the probability that the distance between the actual landing point and the target exceeds 2.5 meters.

(a) 0.249

(b) 0.135

(c) 0.317

(d) 0.05

Question 19. If $X_1, X_2, \ldots, X_n$ are independent exponential random variables with respective rate parameters $\lambda_1, \lambda_2, \ldots, \lambda_n$, which of the following statements is **true** about $Y = \min(X_1, X_2, \ldots, X_n)$?

(a) Y is not exponential for $n > 1$.

(b) Y follows an exponential distribution with parameter $\frac{1}{n}\sum_{i=1}^{n} \lambda_i$.

(c) Y follows an exponential distribution with parameter $\sum_{i=1}^{n} \lambda_i$.

(d) Y follows a gamma distribution with shape n and rate λ_i.

Question 20. For a population with $\mu = 100$, $\sigma^2 = 81$, and sample size $n = 25$, find mean and variance of $\bar{X}$:

(a) Mean = 100, Variance = 81

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(b) **Mean = 100, Variance = 3.24**

(c) Mean = 20, Variance = 81

(d) Mean = 100, Variance = 9

Question 21. Given a population with a mean of $\mu = 100$ and a variance of $\sigma^2 = 81$, the central limit theorem applies when the sample size is $n > 25$. A random sample of size $n = 25$ is obtained. What are the mean and variance of the sampling distribution for the sample means? [Award marks for everyone.]

(a) S^2 always underestimates σ^2

(b) S^2 **is an unbiased estimator of** σ^2**, i.e.,** $E[S^2] = \sigma^2$

(c) S^2 equals σ^2 only when n is large

(d) S^2 estimates the population mean

Question 22. Which of the following statements about the sample variance S^2 is true?

(a) S^2 always underestimates the population variance σ^2.

(b) $\boldsymbol{S^2}$ **is an unbiased estimator of the population variance, i.e.** $\boldsymbol{E[S^2] = \sigma^2}$

(c) S^2 equals the population variance only when the sample size is large.

(d) S^2 estimates the population mean, not the variance.

Question 23. Which of the following best describes the principle of the maximum likelihood estimator (MLE)?

(a) The MLE chooses the parameter values that make the sample variance smallest.

(b) The MLE chooses the parameter values that make the observed data most probable.

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(c) The MLE always equals the sample mean, regardless of the distribution.

(d) The MLE minimizes the expected value of the squared error between the estimate and the true parameter.

Question 24. Let X be a random variable that follows a Gamma distribution with shape parameter $\alpha > 0$ and rate parameter $\lambda > 0$, denoted by

$$X \sim \text{Gamma}(\alpha, \lambda).$$

Which of the following statements is **true** about the Gamma distribution?

(a) The mean and variance of X are $E[X] = \lambda$, $\text{Var}(X) = \alpha$.

(b) The mean and variance of $\boldsymbol{X}$ are $\boldsymbol{E[X] = \frac{\alpha}{\lambda}}$, $\boldsymbol{\text{Var}(X) = \frac{\alpha}{\lambda^2}}$.

(c) The Gamma distribution is always symmetric about its mean.

(d) The probability density function of X is $f(x) = \dfrac{1}{\sqrt{2\pi\lambda}} e^{-\frac{(x-\alpha)^2}{2\lambda}}$.

Subjective Questions

Question 1

There are 100 slips of paper in a hat, each of which has one of the numbers $1, 2, \ldots, 100$ written on it, with no number appearing more than once. Five of the slips are drawn, one at a time. First consider random sampling **with replacement** (with equal probabilities).

(a) What is the distribution of how many of the drawn slips have a value of at least 80 written on them? [0.5 marks]

(b) What is the distribution of the value of the jth draw (for $1 \leq j \leq 5$)? [0.5 marks]

(c) What is the probability that the number 100 is drawn at least once? [1 mark]

Solution:

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(a) Suppose that n independent Bernoulli trials are performed, each with the same success probability p. Let X be the number of successes. The distribution of X is called the Binomial distribution with parameters n and p. By the story of the Binomial, the distribution is

$$\text{Bin}(5, 0.21).$$

(b) Let X_j be the value of the jth draw. By symmetry,

$$X_j \sim \text{DUnif}(1, 2, \ldots, 100).$$

There aren't certain slips that love being chosen on the jth draw and others that avoid being chosen then; all are equally likely.

(c) Taking complements,

$$P(X_j = 100 \text{ for at least one } j) = 1 - P(X_1 \neq 100, \ldots, X_5 \neq 100).$$

By the naive definition of probability, this is

$$1 - \left(\frac{99}{100}\right)^5 \approx 0.049.$$

Question 2
Let X and Y have joint PDF

$$f_{X,Y}(x, y) = cxy, \quad \text{for } 0 < x < y < 1.$$

1. Find c to make this a valid joint PDF. [1 marks]
2. Are X and Y independent? [1 marks]
3. Find the marginal PDFs of X and Y. [1 marks]

Solution:
(1) Find c:

$$\int_0^1 \int_0^y cxy \, dx \, dy = 1$$

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Compute the inner integral:

$$\int_0^y x\,dx = \frac{y^2}{2}$$

So,

$$1 = c\int_0^1 y\cdot\frac{y^2}{2}\,dy = c\int_0^1 \frac{y^3}{2}\,dy = c\cdot\frac{1}{2}\cdot\frac{1}{4} = \frac{c}{8}$$

$$\boxed{c = 8}$$

(2) Check Independence:
For independence, we require

$$f_{X,Y}(x,y) = f_X(x)f_Y(y)$$

for all x, y in the support.

We will compute the marginals below and verify that this condition does not hold.

—

(3) Marginal PDFs:
For X:

$$f_X(x) = \int_{y=x}^1 f_{X,Y}(x,y)\,dy = \int_{y=x}^1 8xy\,dy = 8x\left[\frac{y^2}{2}\right]_{y=x}^1 = 8x\left(\frac{1-x^2}{2}\right)$$

$$\boxed{f_X(x) = 4x(1-x^2)}, \quad 0 < x < 1$$

For Y:

$$f_Y(y) = \int_{x=0}^y f_{X,Y}(x,y)\,dx = \int_{x=0}^y 8xy\,dx = 8y\left[\frac{x^2}{2}\right]_0^y = 8y\cdot\frac{y^2}{2} = 4y^3$$

$$\boxed{f_Y(y) = 4y^3}, \quad 0 < y < 1$$

—

Independence:

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$$f_{X,Y}(x,y) = 8xy, \quad f_X(x)f_Y(y) = [4x(1-x^2)][4y^3] = 16xy^3(1-x^2)$$

These are not equal for all x, y, hence

$$\boxed{X \text{ and } Y \text{ are not independent.}}$$

Question 3

The life of a particular brand of television picture tube is known to be normally distributed with a population standard deviation of $\sigma = 400$ hours. A random sample of $n = 20$ tubes resulted in a sample mean of $\bar{X} = 9000$ hours. Obtain a 90% confidence interval estimate of the mean lifetime of such a tube.(The Z-score that leaves 0.05 in the upper tail is 1.645.) [2 marks]

Solution:

$$\text{Given: } \sigma = 400,\ n = 20,\ \bar{X} = 9000,\ Z_{0.05} = 1.645$$

$$\text{Standard Error (SE)} = \frac{\sigma}{\sqrt{n}} = \frac{400}{\sqrt{20}} = 89.44$$

$$\text{Confidence Interval: } \bar{X} \pm Z_{\alpha/2} \times SE = 9000 \pm 1.645(89.44)$$

$$= 9000 \pm 147.1$$

$$\boxed{(8852.9,\ 9147.1)}$$

Hence, the 90% confidence interval for the mean lifetime is:

$$(8852.9,\ 9147.1)$$

Question 4 To test the hypothesis

$$H_0 : \mu = 105 \quad \text{against} \quad H_1 : \mu \neq 105,$$

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a sample of size $n = 9$ is drawn. If the sample mean is $\bar{X} = 100$, find the p-value when the population standard deviation is known to be 15. [2 marks]

Solution:

$$\mu_0 = 105, \quad \bar{X} = 100, \quad \sigma = 15, \quad n = 9$$

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} = \frac{100 - 105}{15/\sqrt{9}} = \frac{-5}{5} = -1.00$$

$$p = 2P(Z > |z_{\text{obs}}|) = 2P(Z > 1.00)$$

From standard normal tables,

$$P(Z > 1.00) = 0.1587$$

$$\therefore \quad p = 2(0.1587) = 0.3174$$

$$\boxed{p = 0.3174}$$

Question 5 Given a population with mean $\mu = 400$ and variance $\sigma^2 = 1,600$ (Note: $\sigma^2 = 1600$ implied, if $\sigma = 1600$ is intended, the question would be $\sigma = 40$), the central limit theorem applies when the sample size is $n \geq 25$. A random sample of size $n = 35$ is obtained.

(a) What are the mean and variance of the sampling distribution for the sample means? [0.5 marks]

(b) What is the probability that $\bar{x} > 412$? [0.5 marks]

(c) What is the probability that $393 \leq \bar{x} \leq 407$? [1 mark]

(d) What is the probability that $\bar{x} \leq 389$? [0.5 marks]

Solution:

$$\text{Given: } \mu = 400, \ \sigma^2 = 1600 \ (\sigma = 40), \ n = 35.$$

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$$\text{By the CLT, } \bar{X} \approx N\left(\mu,\ \frac{\sigma^2}{n}\right) = N\left(400,\ \frac{1600}{35}\right).$$

(a) Mean and variance of the sampling distribution:

$$E(\bar{X}) = 400, \qquad \text{Var}(\bar{X}) = \frac{1600}{35} \approx 45.7143,$$

and the standard error is

$$\sigma_{\bar{X}} = \sqrt{\frac{1600}{35}} \approx 6.7612.$$

(b) $P(\bar{X} > 412)$.

$$z = \frac{412 - 400}{\sigma_{\bar{X}}} = \frac{12}{6.7612} \approx 1.7748,$$

$$P(\bar{X} > 412) = 1 - \Phi(1.7748) \approx 0.03796.$$

(c) $P(393 \leq \bar{X} \leq 407)$.

$$z_1 = \frac{393 - 400}{6.7612} \approx -1.0353, \qquad z_2 = \frac{407 - 400}{6.7612} \approx 1.0353,$$

$$P(393 \leq \bar{X} \leq 407) = \Phi(1.0353) - \Phi(-1.0353) \approx 0.69948.$$

(d) $P(\bar{X} \leq 389)$.

$$z = \frac{389 - 400}{6.7612} \approx -1.6269,$$

$$P(\bar{X} \leq 389) = \Phi(-1.6269) \approx 0.05188.$$

$\boxed{\begin{array}{l}(a)\ E(\bar{X}) = 400,\ \text{Var}(\bar{X}) = \frac{1600}{35} \approx 45.7143, \\ (b)\ P(\bar{X} > 412) \approx 0.03796, \\ (c)\ P(393 \leq \bar{X} \leq 407) \approx 0.69948, \\ (d)\ P(\bar{X} \leq 389) \approx 0.05188.\end{array}}$

Question 6 A firm employs 189 junior accountants. In a random sample of 50 of these, the mean number of hours over-time billed in a particular week was 9.7, and the sample standard deviation was 6.2 hours.

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(a) Find a 95% confidence interval for the mean number of hours overtime billed per junior accountant in this firm that week. [1 mark]

(b) Find a 99% confidence interval for the total number of hours overtime billed by junior accountants in the firm during the week of interest. [1 mark]

Solution:

$$\text{Given: } n = 50,\ \bar{x} = 9.7,\ s = 6.2,\ df = n - 1 = 49.$$

$$\text{Standard error: } \mathrm{SE}(\bar{X}) = \frac{s}{\sqrt{n}} = \frac{6.2}{\sqrt{50}} \approx 0.8768124.$$

$$t_{0.025,49} \approx 2.0096, \qquad t_{0.005,49} \approx 2.678.$$

(a) 95% CI for the mean:

$$\bar{x} \pm t_{0.975,49}\,\mathrm{SE}(\bar{X}) = 9.7 \pm 2.0096(0.8768124) = 9.7 \pm 1.7620422,$$

$$\boxed{95\%\text{ CI for } \mu:\ (7.93796,\ 11.46204)}$$

(b) 99% CI for the total hours (first find 99% CI for the mean):

$$9.7 \pm t_{0.995,49}\,\mathrm{SE}(\bar{X}) = 9.7 \pm 2.678(0.8768124) = 9.7 \pm 2.3481036,$$

so the 99% CI for the mean is $(7.35190,\ 12.04810)$.

Multiplying by $N = 189$ gives the CI for the total:

$$\boxed{(1389.51,\ 2277.09)}$$

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